

Evidencias de Entrega — API de Categorías con FastAPI

Nombre del aprendiz: Sara Jasmin Cruz Calderon

Ficha: 3407180

1. Captura de Swagger (/docs)

Vista general — Endpoints disponibles

Documentación interactiva de Swagger UI generada por FastAPI, con el listado completo de endpoints de productos y categorías.

default		^
GET	/ Root	▼
GET	/health Health	▼
GET	/products/ Get Products	▼ 
POST	/products/ Create Product	▼
GET	/products/{product_id} Get Product By Id	▼
PATCH	/products/{product_id} Update Product	▼
DELETE	/products/{product_id} Delete Product	▼
GET	/categories Get Categories	▼
POST	/categories Create Category	▼
GET	/categories/{category_id} Get Category	▼
PATCH	/categories/{category_id} Update Category	▼
DELETE	/categories/{category_id} Delete Category	▼
Schemas		^

CA01 — Listar categorías

GET /categories · Esperado: 200 y lista JSON. La respuesta muestra el arreglo completo de categorías registradas.

GET /categories Get Categories

Parameters

Cancel

Name	Description
active	--
boolean (boolean null) (query)	
search	search
string (string null) (query)	

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/categories' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/categories
```

Server response

Code

Details

200

Response body

```
{
  {
    "name": "Computadores",
    "description": "Equipos de cómputo",
    "active": true,
    "id": 1
  },
  {
    "name": "Accesorios",
    "description": "Periféricos y accesorios",
    "active": false,
```

CA02 — Consultar categoría existente *GET /categories/{category_id} · Esperado: 200. Se consulta una categoría existente y se obtiene su detalle compl*

GET /categories/{category_id} Get Category

Parameters

Cancel

Name	Description
category_id * required integer (path)	2

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/categories/2' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/categories/2
```

Server response

Code	Details
200	Response body <pre>{ "name": "Accesorios", "description": "Periféricos y accesorios", "active": false, "id": 2 }</pre> Download Response headers <pre>content-length: 85 content-type: application/json date: Fri, 11 Sep 2026 21:09:02 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

CA03 — Consultar categoría inexistente

GET /categories/56 · Esperado: 404.

GET

/categories/{category_id} Get Category

⌵

Parameters

Cancel

Name	Description
category_id * required	
integer (path)	56

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/categories/56' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/categories/56
```

Server response

Code	Details
404	Error: Not Found

Undocumented

Response body

```
{
  "detail": "Category with id 56 not found"
}
```

Download

Response headers

```
content-length: 42
content-type: application/json
date: Fri, 11 Sep 2026 21:34:47 GMT
server: uvicorn
```

Responses

Code	Description	Links
------	-------------	-------

CA04 — ID inválido

GET /categories/abc · Esperado: 422.

GET /categories/{category_id} Get Category

Parameters Cancel

Name	Description
category_id * required	
integer (path)	abc

Please correct the following validation errors and try again.

- For 'category_id': Value must be an integer.

Execute

```
INFO:      127.0.0.1:57639 - "GET /categories/abc HTTP/
1.1" 422 Unprocessable Content
INFO:      127.0.0.1:57771 - "GET /categories/abc HTTP/1.1" 422 U
nprocessable Content
█
```

CA05 — Crear categoría válida

POST /categories con datos correctos · Esperado: 201. La categoría se crea correctamente y se retorna con su id asignado.

POST

/categories

Create Category

^

Parameters

Cancel

No parameters

Request body

required

application/json

▼

Edit Value

Schema

```
{
  "name": "Computadores",
  "description": "Equipos de cómputo",
  "active": true
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/categories' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Computadores",
    "description": "Equipos de cómputo",
    "active": true
  }'
```

Request URL

http://127.0.0.1:8000/categories

Server response

Code

Details

201

Response body

```
{
  "name": "Computadores",
  "description": "Equipos de cómputo",
  "active": true,
  "id": 1
}
```

Download

CA06 — Nombre demasiado corto

POST /categories con name de 1-2 caracteres · Esperado: 422.

POST /categories Create Category

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{  "name": "S3",  "description": "Equipos de cómputo",  "active": true}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/categories' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "S3",
    "description": "Equipos de cómputo",
    "active": true
  }'
```

Request URL

http://127.0.0.1:8000/categories

Server response

Code	Details
422	Error: Unprocessable Content Response body <pre>{ "detail": [{ "type": "string_too_short", "loc": ["body", "name"], "msg": "String should have at least 3 characters", "input": "S3", "ctx": { "min_length": 3 } }]}</pre> Response headers <pre>content-length: 147 content-type: application/json date: Fri, 11 Sep 2026 21:57:24 GMT server: uvicorn</pre>

Responses

Code	Description	Links
------	-------------	-------

CA07 — Falta nombre

POST /categories sin name · Esperado: 422.

POST

/categories

Create Category

^

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{  "name": "",  "description": "Equipos de cómputo",  "active": true}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/categories' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "name": "",
    "description": "Equipos de cómputo",
    "active": true
  }'
```

Request URL

http://127.0.0.1:8000/categories

Server response

Code

Details

422

Error: Unprocessable Content

Response body

```
{  "detail": [    {      "type": "string_too_short",      "loc": [        "body",        "name"      ],      "msg": "String should have at least 3 characters",      "input": "",      "ctx": {        "min_length": 3      }    }  ]}
```

Download

Response headers

```
content-length: 145
content-type: application/json
date: Fri, 11 Sep 2026 22:09:00 GMT
server: uvicorn
```

Responses

CA08 — Actualizar categoría existente

PATCH /categories/{category_id} · Esperado: 200. La categoría existente se actualiza correctamente con los nuevos datos.

PATCH /categories/{category_id} Update Category

Parameters

Cancel

Name	Description
category_id * required	
integer	1
(path)	

Request body required

application/json

Edit Value | Schema

```
{
  "name": "string",
  "description": "string",
  "active": true
}
```

Execute

Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8000/categories/1' \
  -H 'accept: application/json' \
  -H 'Content-type: application/json' \
  -d '{
    "name": "string",
    "description": "string",
    "active": true
  }'
```

Request URL

http://127.0.0.1:8000/categories/1

Server response

Code	Details
200	Response body <pre>{ "name": "string", "description": "string", "active": true, "id": 1 }</pre> Download Response headers <pre>content-length: 61 content-type: application/json date: Fri, 11 Sep 2026 21:18:01 GMT server: uvicorn</pre>

CA09 — Actualizar categoría inexistente

PATCH /categories/999 · Esperado: 404.

PATCH /categories/{category_id} Update Category

Parameters

Name

Description

category_id required

integer

(path)

999

Request body required

application/json

Edit Value | Schema

{
 "name": "string",
 "description": "string",
 "active": true
}

Execute

Clear

Responses

Curl

```
curl -X 'PATCH' \  
  'http://127.0.0.1:8000/categories/999' \  
  -H 'accept: application/json' \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "name": "string",  
    "description": "string",  
    "active": true  
  }'
```

Request URL

http://127.0.0.1:8000/categories/999

Server response

Code

Details

404

Error: Not Found

Response body

{
 "detail": "Category with id 999 not found"
}

Download

Response headers

content-length: 43
content-type: application/json
date: Fri, 11 Sep 2026 22:13:05 GMT
server: uvicorn

Responses

Code

Description

Links

CA10 — Eliminar categoría existente

DELETE /categories/{category_id} · Esperado: 204. La categoría se elimina correctamente, sin contenido en la respuesta.

DELETE /categories/{category_id} Delete Category

Parameters

Name	Description
category_id required	
integer	
(path)	

1

Execute

Clear

Responses

Curl

curl -X 'DELETE' \n'http://127.0.0.1:8000/categories/1' \n-H 'accept: */*'

Request URL

http://127.0.0.1:8000/categories/1

Server response

Code	Details
204	Response headers date: Fri, 11 Sep 2026 21:11:38 GMT\nserver: uvicorn

Responses

Code	Description	Links
204	Successful Response	No links
422	Validation Error	No links

Media type

application/json

Example Value

Schema

{\n \"detail\": [\n {\n \"loc\": [\n \"string\",\n 0\n],\n \"msg\": \"string\",\n \"type\": \"string\",\n \"input\": \"string\",\n \"ctx\": {}\n }\n]\n}

CA11 — Eliminar categoría inexistente

DELETE /categories/999 · Esperado: 404.

DELETE /categories/{category_id} Delete Category

Parameters

Name	Description
category_id * required	
integer	999
(path)	

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/categories/999' \
-H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/categories/999
```

Server response

Code	Details
404	Error: Not Found

Response body

```
{
  "detail": "Category with id 999 not found"
}
```

Download

Response headers

```
content-length: 43
content-type: application/json
date: Fri, 11 Sep 2026 22:16:43 GMT
server: uvicorn
```

Responses

CA12 — Filtrar categorías activas

GET /categories?active=true · Esperado: 200 y solo resultados con active=true.

GET /categories Get Categories

Parameters

Cancel

Name	Description
active	—
boolean (boolean null) (query)	
search	Computadores
string (string null) (query)	

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/categories?search=Computadores' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/categories?search=Computadores
```

Server response

Code	Details
200	Response body <pre>{ "name": "Computadores", "description": "Equipos de cómputo", "active": true, "id": 1 }</pre> Download Response headers <pre>content-length: 82 content-type: application/json date: Fri, 11 Sep 2026 22:28:43 GMT server: uvicorn</pre>

GET/categoriesGet Categories

Parameters

active

boolean | (boolean | null) (query)

false

search

string | (string | null) (query)

search

Execute

Clear

Responses

Curl

curl -X 'GET' \ http://127.0.0.1:8000/categories?active=false \ -H 'accept: application/json'

Request URL

http://127.0.0.1:8000/categories?active=false

Server response

Code

Details

200

Response body

[{"name": "Accesorios", "description": "Perif\u00e9ricos y accesorios", "active": false, "id": 2}]

Download

Response headers

content-length: 87 content-type: application/json date: Fri, 11 Sep 2026 22:21:17 GMT server: uvicorn

Responses

2. Captura de ejecuci\u00f3n de pruebas (pytest -v)

```
(.venv) PS C:\Users\moise\OneDrive\Desktop\API-DE-CATEGOR-AS-Y-PRODUCTOS-CON-FASTAPI-SARA-CRUZ> pytest
===== test session starts =====
platform win32 -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\moise\OneDrive\Desktop\API-DE-CATEGOR-AS-Y-PRODUCTOS-CON-FASTAPI-SARA-CRUZ\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\moise\OneDrive\Desktop\API-DE-CATEGOR-AS-Y-PRODUCTOS-CON-FASTAPI-SARA-CRUZ
plugins: anyio-4.15.1
collected 23 items

tests/test_categories.py::test_list_categories PASSED [ 4%]
tests/test_categories.py::test_get_existing_category PASSED [ 8%]
tests/test_categories.py::test_get_non_existing_category PASSED [ 13%]
tests/test_categories.py::test_get_invalid_id PASSED [ 17%]
tests/test_categories.py::test_create_valid_category PASSED [ 21%]
tests/test_categories.py::test_create_category_name_too_short PASSED [ 26%]
tests/test_categories.py::test_create_category_missing_name PASSED [ 30%]
tests/test_categories.py::test_update_existing_category PASSED [ 34%]
tests/test_categories.py::test_update_non_existing_category PASSED [ 39%]
tests/test_categories.py::test_delete_existing_category PASSED [ 43%]
tests/test_categories.py::test_delete_non_existing_category PASSED [ 47%]
tests/test_categories.py::test_filter_active_categories PASSED [ 52%]
tests/test_categories.py::test_search_categories PASSED [ 56%]
tests/test_categories.py::test_search_categories_no_match PASSED [ 60%]
tests/test_products.py::test_health PASSED [ 65%]
tests/test_products.py::test_get_products PASSED [ 69%]
tests/test_products.py::test_get_existing_product PASSED [ 73%]
tests/test_products.py::test_get_non_existing_product PASSED [ 78%]
tests/test_products.py::test_create_product PASSED [ 82%]
tests/test_products.py::test_update_product PASSED [ 86%]
tests/test_products.py::test_update_product_stock_rec calculates available PASSED [ 91%]
tests/test_products.py::test_delete_product PASSED [ 95%]
tests/test_products.py::test_filters PASSED [100%]

===== warnings summary =====
..venv\Lib\site-packages\starlette\testclient.py:53
  C:\Users\moise\OneDrive\Desktop\API-DE-CATEGOR-AS-Y-PRODUCTOS-CON-FASTAPI-SARA-CRUZ\.venv\Lib\site-packages\starlette\testclient.py:53: DeprecationWarning
: The anyio.bcp.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
  _PortalFactoryType = Callable[[], AbstractContextManager[anyio.bcp.BlockingPortal]]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 23 passed, 1 warning in 0.22s =====
(.venv) PS C:\Users\moise\OneDrive\Desktop\API-DE-CATEGOR-AS-Y-PRODUCTOS-CON-FASTAPI-SARA-CRUZ>
```

3. Matriz de casos de prueba

Resultado obtenido para cada caso de prueba: PASS o FAIL.

ID	Escenario	Petición / condición	Esperado	Resultado obtenido
CA01	Listar categorías	GET /categories	200 y lista JSON	200 – PASS
CA02	Consultar existente	GET /categories/1	200	200 – PASS
CA03	Consultar inexistente	GET /categories/56	404	404 – PASS
CA04	ID inválido	GET /categories/abc	422	422 – PASS
CA05	Crear válida	POST con datos correctos	201	201 – PASS
CA06	Nombre demasiado corto	POST con name de 1-2 caracteres	422	422 – PASS
CA07	Falta nombre	POST sin name	422	422 – PASS
CA08	Actualizar existente	PATCH /categories/1	200	200 – PASS
CA09	Actualizar inexistente	PATCH /categories/999	404	404 – PASS
CA10	Eliminar existente	DELETE /categories/1	204	204 – PASS
CA11	Eliminar inexistente	DELETE /categories/999	404	404 – PASS
CA12	Filtrar activas	GET /categories?active=true	200 y solo active=true	200 – PASS

4. Observaciones adicionales

(Espacio libre para comentarios sobre incidencias, decisiones de diseño o casos adicionales implementados, como el reto opcional de búsqueda por nombre.)

*Además de lo pedido en la guía, agregué un filtro de búsqueda por nombre en el endpoint de categorías (parámetro **search**), y también dejé la opción de filtrar solo las categorías activas con **active=true**. Esto no afecta el funcionamiento normal del endpoint, es opcional.*

En el caso CA04 (cuando se manda un id que no es número, como "abc"), me di cuenta de que Swagger no deja ver el código HTTP porque él mismo bloquea la petición antes de mandarla al servidor; ya que valida que el dato sea numérico. Para comprobar que la API sí responde bien, probé la misma petición desde la terminal con curl y ahí sí me apareció el código 422 con el mensaje de error; tal como se esperaba.

Corrí toda la suite de pruebas con `pytest -v` y todos los casos pasaron sin errores.